

PRAKTIKUM SISTEM OPERASI

MODUL XV

KEAMANAN



Oleh:

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2311104053

PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

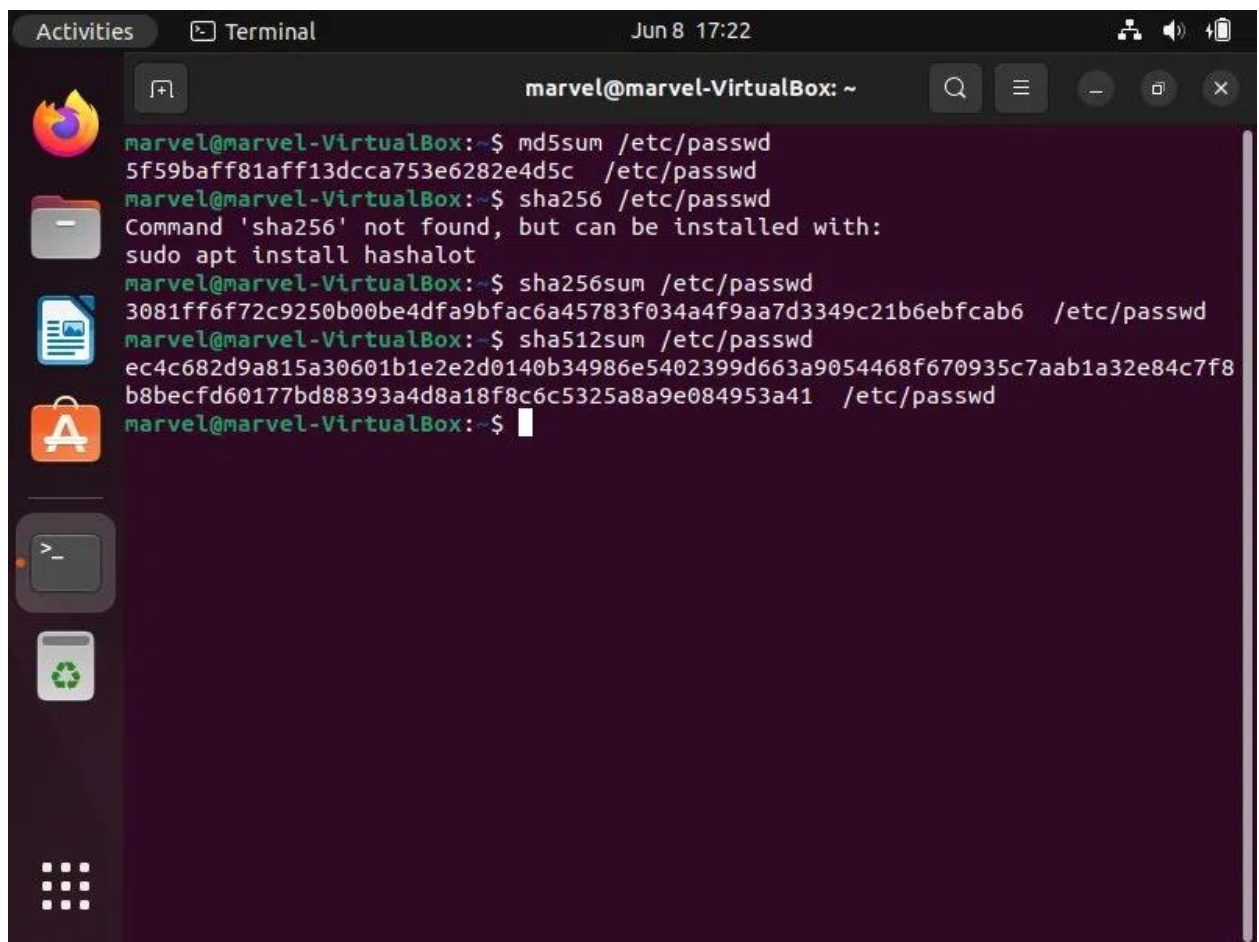
2026

JURNAL

1. Integritas: Dasar Hashing

- a. Hash file /etc/passwd
- b. Buat file test_0.txt berisi isi /etc/passwd
- c. Hash file test_0.txt
- d. Rename lalu hash lagi
- e. Analisis

Jawaban:



```
marvel@marvel-VirtualBox: ~  
$ md5sum /etc/passwd  
5f59baff81aff13dcca753e6282e4d5c /etc/passwd  
$ sha256 /etc/passwd  
Command 'sha256' not found, but can be installed with:  
sudo apt install hashalot  
$ sha256sum /etc/passwd  
3081ff6f72c9250b00be4dfa9bfac6a45783f034a4f9aa7d3349c21b6ebfcab6 /etc/passwd  
$ sha512sum /etc/passwd  
ec4c682d9a815a30601b1e2e2d0140b34986e5402399d663a9054468f670935c7aab1a32e84c7f8  
b8becfd60177bd88393a4d8a18f8c6c5325a8a9e084953a41 /etc/passwd  
$
```

Activities Terminal Jun 8 17:23

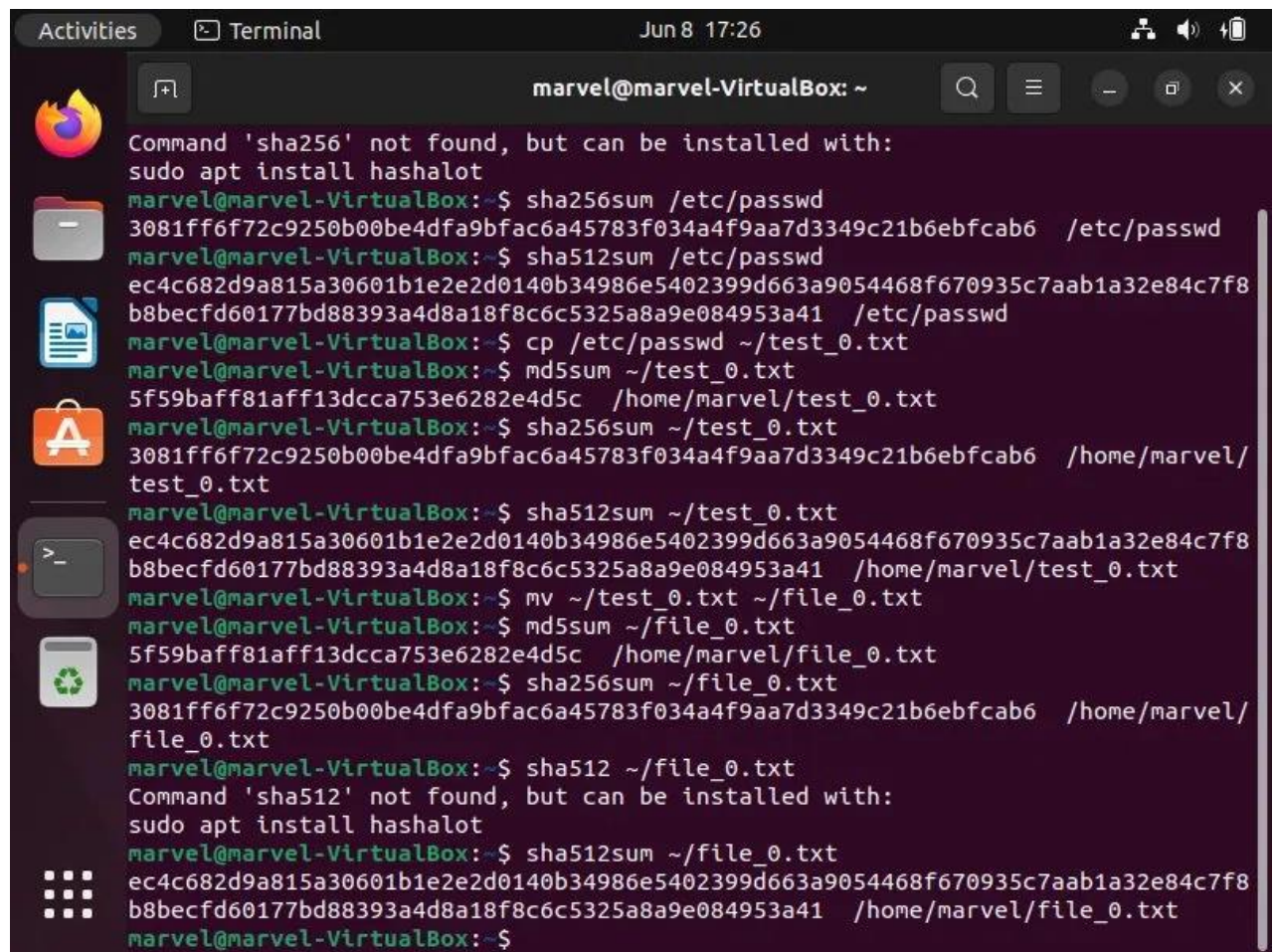
marvel@marvel-VirtualBox: ~

```
marvel@marvel-VirtualBox:~$ md5sum /etc/passwd
5f59baff81aff13dcca753e6282e4d5c /etc/passwd
marvel@marvel-VirtualBox:~$ sha256 /etc/passwd
Command 'sha256' not found, but can be installed with:
sudo apt install hashalot
marvel@marvel-VirtualBox:~$ sha256sum /etc/passwd
3081ff6f72c9250b00be4dfa9bfac6a45783f034a4f9aa7d3349c21b6ebfcab6 /etc/passwd
marvel@marvel-VirtualBox:~$ sha512sum /etc/passwd
ec4c682d9a815a30601b1e2e2d0140b34986e5402399d663a9054468f670935c7aab1a32e84c7f8
b8becfd60177bd88393a4d8a18f8c6c5325a8a9e084953a41 /etc/passwd
marvel@marvel-VirtualBox:~$ cp /etc/passwd ~/test_0.txt
marvel@marvel-VirtualBox:~$
```

Activities Terminal Jun 8 17:25

marvel@marvel-VirtualBox: ~

```
marvel@marvel-VirtualBox:~$ md5sum /etc/passwd
5f59baff81aff13dcca753e6282e4d5c /etc/passwd
marvel@marvel-VirtualBox:~$ sha256 /etc/passwd
Command 'sha256' not found, but can be installed with:
sudo apt install hashalot
marvel@marvel-VirtualBox:~$ sha256sum /etc/passwd
3081ff6f72c9250b00be4dfa9bfac6a45783f034a4f9aa7d3349c21b6ebfcab6 /etc/passwd
marvel@marvel-VirtualBox:~$ sha512sum /etc/passwd
ec4c682d9a815a30601b1e2e2d0140b34986e5402399d663a9054468f670935c7aab1a32e84c7f8
b8becfd60177bd88393a4d8a18f8c6c5325a8a9e084953a41 /etc/passwd
marvel@marvel-VirtualBox:~$ cp /etc/passwd ~/test_0.txt
marvel@marvel-VirtualBox:~$ md5sum ~/test_0.txt
5f59baff81aff13dcca753e6282e4d5c /home/marvel/test_0.txt
marvel@marvel-VirtualBox:~$ sha256sum ~/test_0.txt
3081ff6f72c9250b00be4dfa9bfac6a45783f034a4f9aa7d3349c21b6ebfcab6 /home/marvel/
test_0.txt
marvel@marvel-VirtualBox:~$ sha512sum ~/test_0.txt
ec4c682d9a815a30601b1e2e2d0140b34986e5402399d663a9054468f670935c7aab1a32e84c7f8
b8becfd60177bd88393a4d8a18f8c6c5325a8a9e084953a41 /home/marvel/test_0.txt
marvel@marvel-VirtualBox:~$
```



```
Activities Terminal Jun 8 17:26
marvel@marvel-VirtualBox: ~
Command 'sha256' not found, but can be installed with:
sudo apt install hashalot
marvel@marvel-VirtualBox:~$ sha256sum /etc/passwd
3081ff6f72c9250b00be4dfa9bfac6a45783f034a4f9aa7d3349c21b6ebfcab6 /etc/passwd
marvel@marvel-VirtualBox:~$ sha512sum /etc/passwd
ec4c682d9a815a30601b1e2e2d0140b34986e5402399d663a9054468f670935c7aab1a32e84c7f8
b8becfd60177bd88393a4d8a18f8c6c5325a8a9e084953a41 /etc/passwd
marvel@marvel-VirtualBox:~$ cp /etc/passwd ~/test_0.txt
marvel@marvel-VirtualBox:~$ md5sum ~/test_0.txt
5f59baff81aff13dcca753e6282e4d5c /home/marvel/test_0.txt
marvel@marvel-VirtualBox:~$ sha256sum ~/test_0.txt
3081ff6f72c9250b00be4dfa9bfac6a45783f034a4f9aa7d3349c21b6ebfcab6 /home/marvel/
test_0.txt
marvel@marvel-VirtualBox:~$ sha512sum ~/test_0.txt
ec4c682d9a815a30601b1e2e2d0140b34986e5402399d663a9054468f670935c7aab1a32e84c7f8
b8becfd60177bd88393a4d8a18f8c6c5325a8a9e084953a41 /home/marvel/test_0.txt
marvel@marvel-VirtualBox:~$ mv ~/test_0.txt ~/file_0.txt
marvel@marvel-VirtualBox:~$ md5sum ~/file_0.txt
5f59baff81aff13dcca753e6282e4d5c /home/marvel/file_0.txt
marvel@marvel-VirtualBox:~$ sha256sum ~/file_0.txt
3081ff6f72c9250b00be4dfa9bfac6a45783f034a4f9aa7d3349c21b6ebfcab6 /home/marvel/
file_0.txt
marvel@marvel-VirtualBox:~$ sha512 ~/file_0.txt
Command 'sha512' not found, but can be installed with:
sudo apt install hashalot
marvel@marvel-VirtualBox:~$ sha512sum ~/file_0.txt
ec4c682d9a815a30601b1e2e2d0140b34986e5402399d663a9054468f670935c7aab1a32e84c7f8
b8becfd60177bd88393a4d8a18f8c6c5325a8a9e084953a41 /home/marvel/file_0.txt
marvel@marvel-VirtualBox:~$
```

Langkah-langkah:

a. Hash file /etc/passwd

Jalankan 3 perintah ini satu per satu di terminal Linux:

`md5sum /etc/passwd`

`sha256sum /etc/passwd`

`sha512sum /etc/passwd`

b. Buat file test_0.txt berisi isi /etc/passwd

`cp /etc/passwd ~/test_0.txt`

c. Hash file test_0.txt

`md5sum ~/test_0.txt`

`sha256sum ~/test_0.txt`

`sha512sum ~/test_0.txt`

d. Rename lalu hash lagi

`mv ~/test_0.txt ~/file_0.txt`


```
md5sum ~/file_0.txt
sha256sum ~/file_0.txt
sha512sum ~/file_0.txt
```

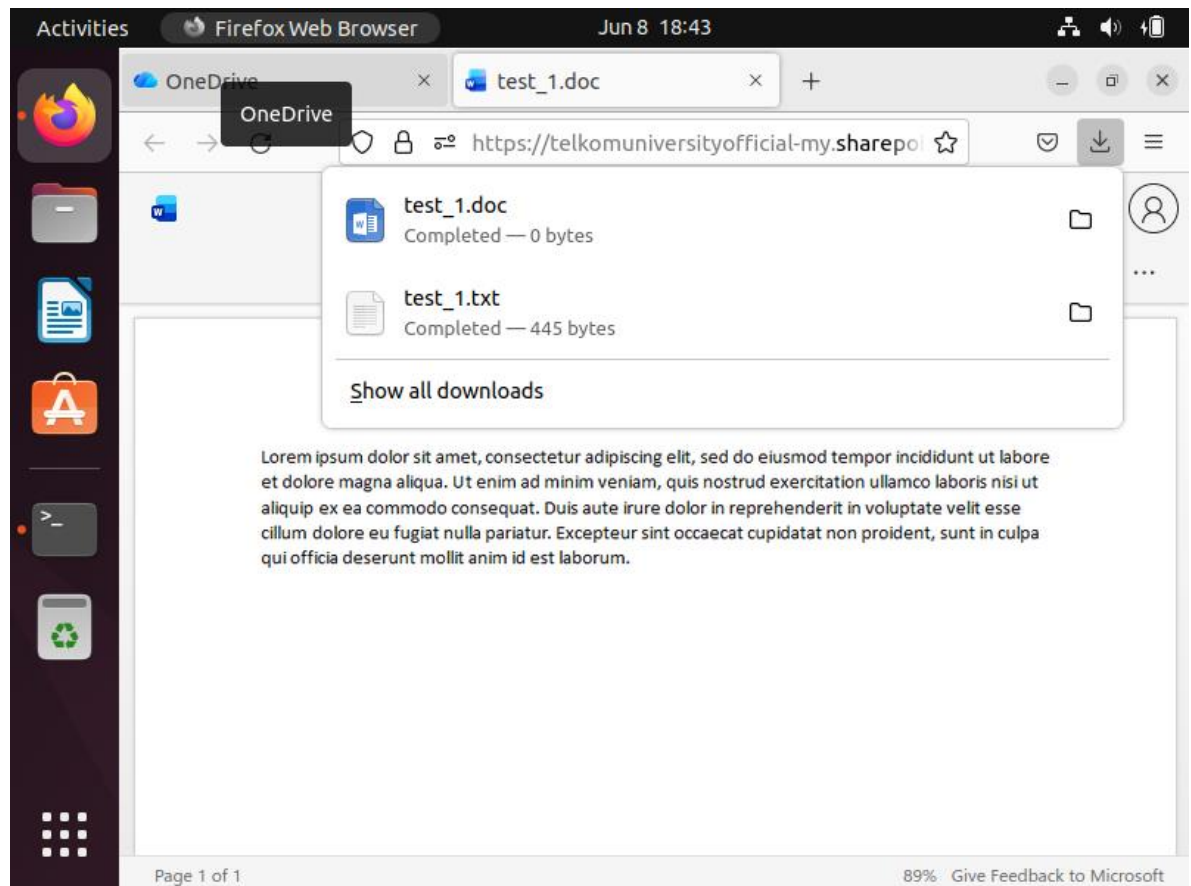
e. Analisis

Ketiga file (/etc/passwd, test_0.txt, file_0.txt) memiliki hash yang sama persis. Karena isinya identik — rename tidak mengubah konten file, dan cp hanya menduplikasi isi. Hash hanya bergantung pada isi/konten, bukan nama file.

2. Avalanche (test_1.txt)

- Download file bernama test_1.txt
- Hash test_1.txt sebelum diubah
- Hapus titik di akhir file
- Hash lagi setelah diubah
- Analisis

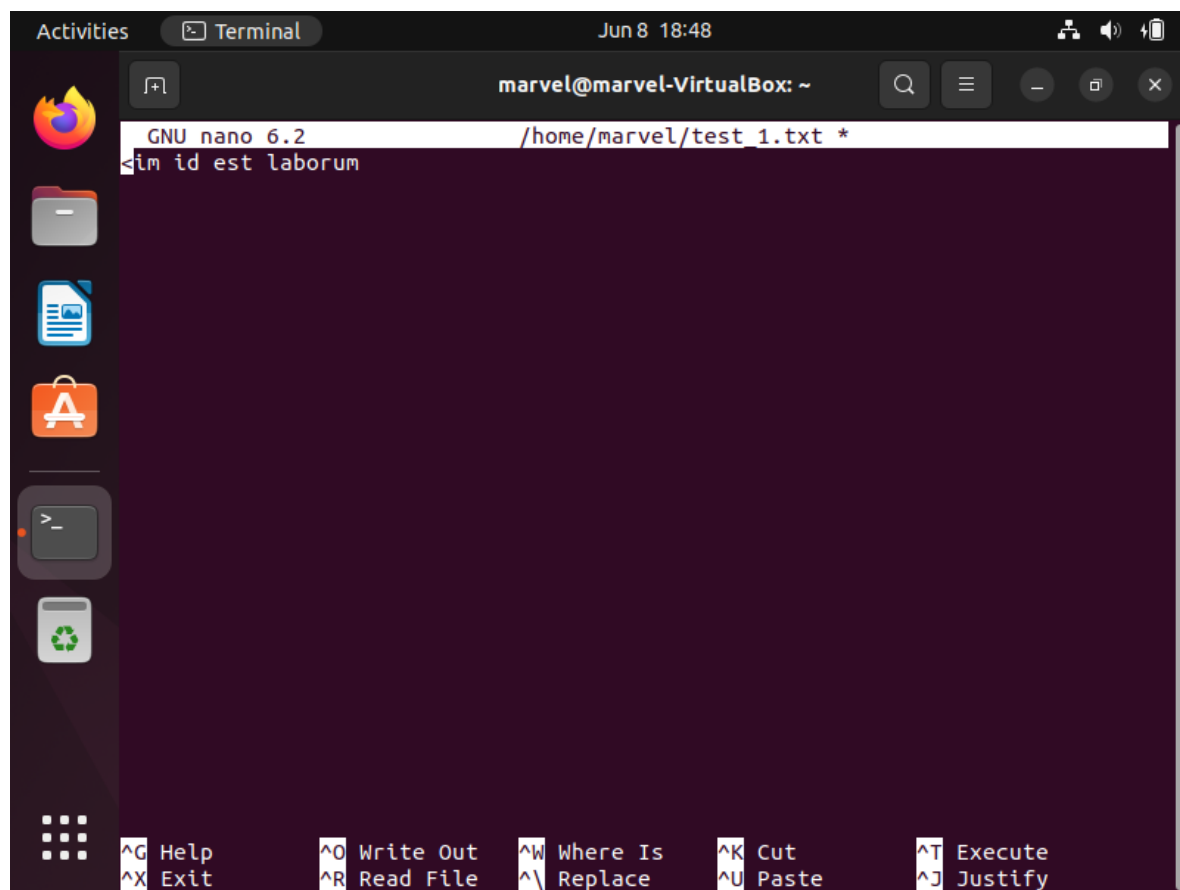
Jawaban:

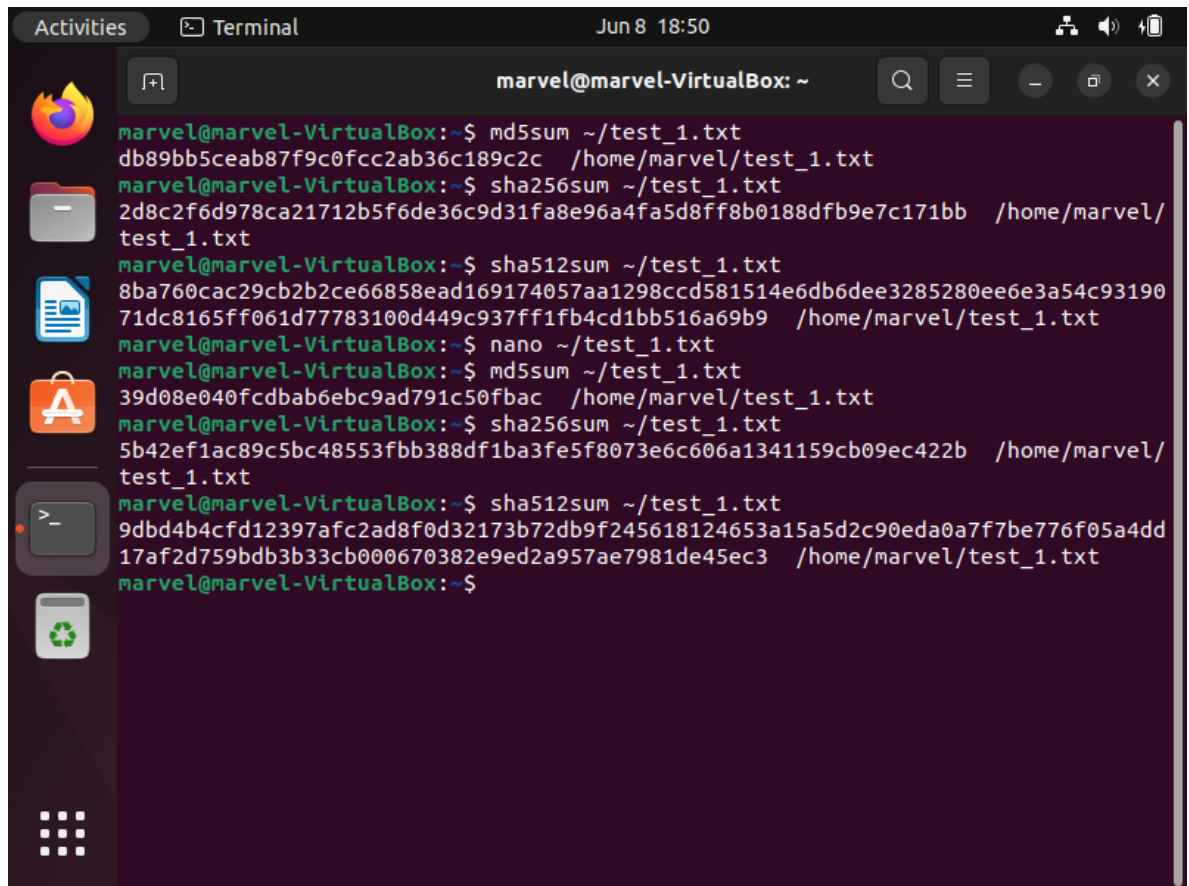


Activities Terminal Jun 8 18:47

marvel@marvel-VirtualBox: ~

```
marvel@marvel-VirtualBox:~$ md5sum ~/test_1.txt
db89bb5ceab87f9c0fcc2ab36c189c2c  /home/marvel/test_1.txt
marvel@marvel-VirtualBox:~$ sha256sum ~/test_1.txt
2d8c2f6d978ca21712b5f6de36c9d31fa8e96a4fa5d8ff8b0188dfb9e7c171bb  /home/marvel/
test_1.txt
marvel@marvel-VirtualBox:~$ sha512sum ~/test_1.txt
8ba760cac29cb2b2ce66858ead169174057aa1298ccd581514e6db6dee3285280ee6e3a54c93190
71dc8165ff061d77783100d449c937ff1fb4cd1bb516a69b9  /home/marvel/test_1.txt
marvel@marvel-VirtualBox:~$
```





```
marvel@marvel-VirtualBox:~$ md5sum ~/test_1.txt
db89bb5ceab87f9c0fcc2ab36c189c2c  /home/marvel/test_1.txt
marvel@marvel-VirtualBox:~$ sha256sum ~/test_1.txt
2d8c2f6d978ca21712b5f6de36c9d31fa8e96a4fa5d8ff8b0188dfb9e7c171bb  /home/marvel/
test_1.txt
marvel@marvel-VirtualBox:~$ sha512sum ~/test_1.txt
8ba760cac29cb2b2ce66858ead169174057aa1298ccd581514e6db6dee3285280ee6e3a54c93190
71dc8165ff061d77783100d449c937ff1fb4cd1bb516a69b9  /home/marvel/test_1.txt
marvel@marvel-VirtualBox:~$ nano ~/test_1.txt
marvel@marvel-VirtualBox:~$ md5sum ~/test_1.txt
39d08e040fcdab6ebc9ad791c50fbac  /home/marvel/test_1.txt
marvel@marvel-VirtualBox:~$ sha256sum ~/test_1.txt
5b42ef1ac89c5bc48553fbb388df1ba3fe5f8073e6c606a1341159cb09ec422b  /home/marvel/
test_1.txt
marvel@marvel-VirtualBox:~$ sha512sum ~/test_1.txt
9dbd4b4cfd12397afc2ad8f0d32173b72db9f245618124653a15a5d2c90eda0a7f7be776f05a4dd
17af2d759bdb3b33cb000670382e9ed2a957ae7981de45ec3  /home/marvel/test_1.txt
marvel@marvel-VirtualBox:~$
```

Langkah-langkah:

- Download file test_1.txt
wget https://tiny.cc/test1_txt -o test_1.txt
- Hash test_1.txt sebelum diubah
md5sum ~/test_1.txt
sha256sum ~/test_1.txt
sha512sum ~/test_1.txt
- Hapus titik di akhir file
tail ~/test_1.txt
Lalu buka filenya dengan text editor:
nano ~/test_1.txt

Pakai tombol End untuk ke akhir baris terakhir

Hapus tanda titik (.) di paling akhir

Ctrl+X → tekan Y → Enter untuk save

- Hash lagi setelah diubah
md5sum ~/test_1.txt
sha256sum ~/test_1.txt

```
sha512sum ~/test_1.txt
```

e. Analisis

Nilai hash berubah total meskipun hanya menghapus 1 karakter titik. Inilah efek Avalanche — perubahan sekecil apapun pada input menghasilkan hash yang sama sekali berbeda. Ini membuktikan hash sensitif terhadap perubahan sekecil apapun.

3. Metadata (test_1.doc)

a. Download file bernama test_1.doc

b. Hash test_1.doc sebelum diubah

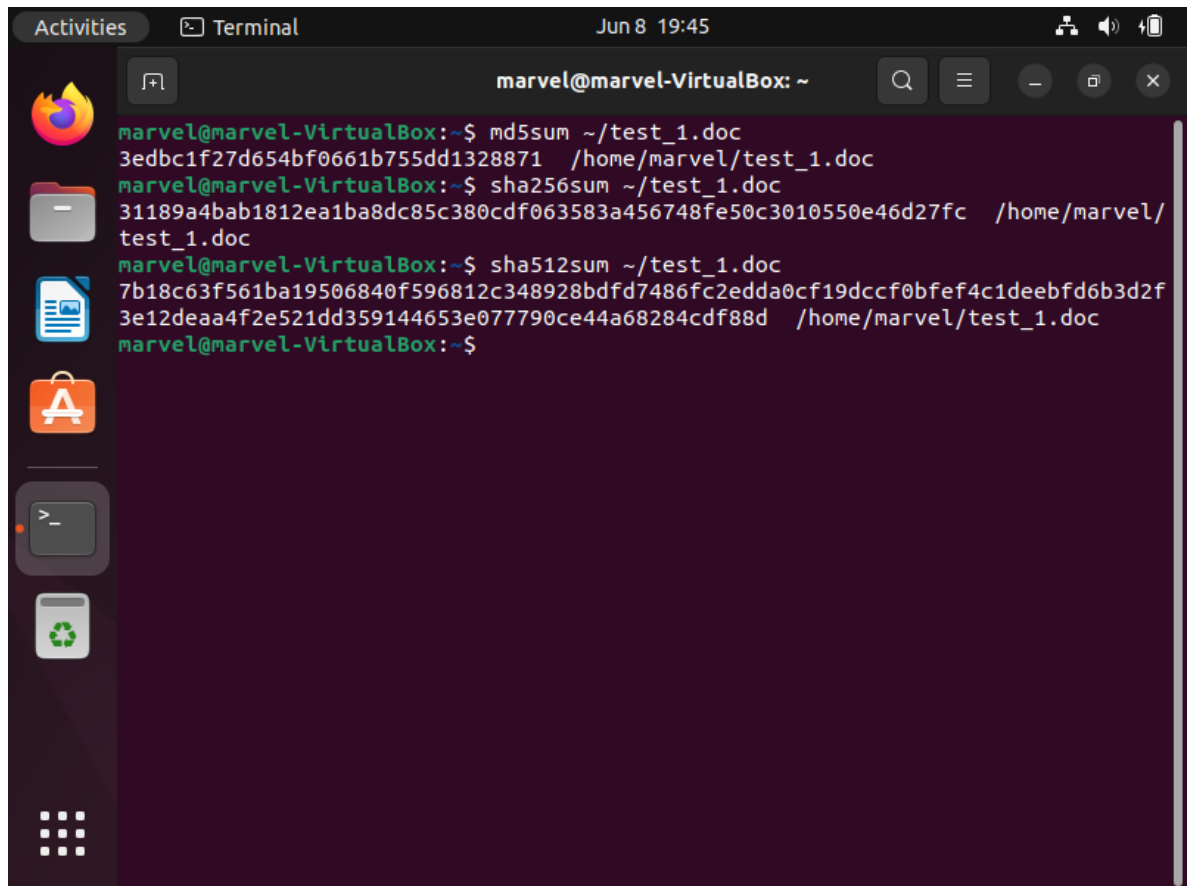
c. i. Buka lagi, ketik abcdef, save

ii. Hapus abcdef, save lagi

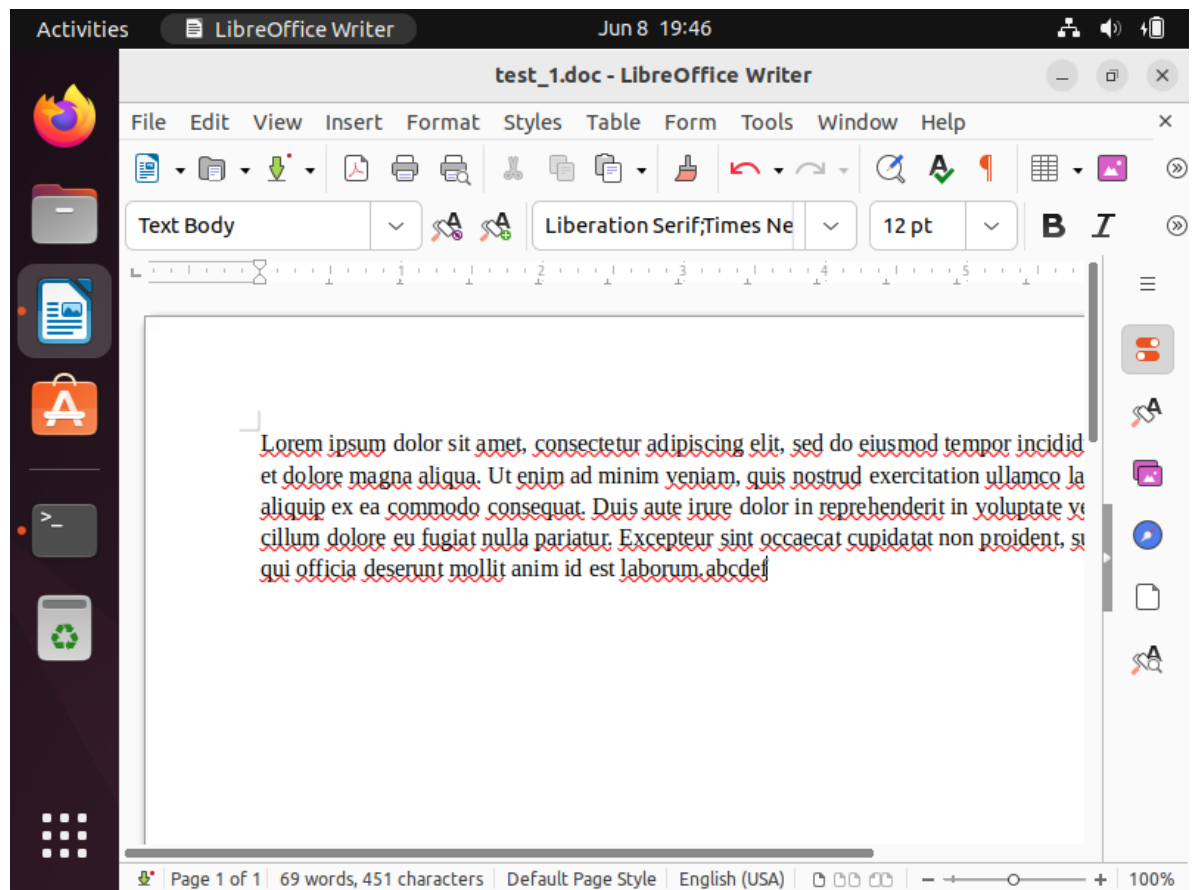
d. Hash lagi setelah semua perubahan.

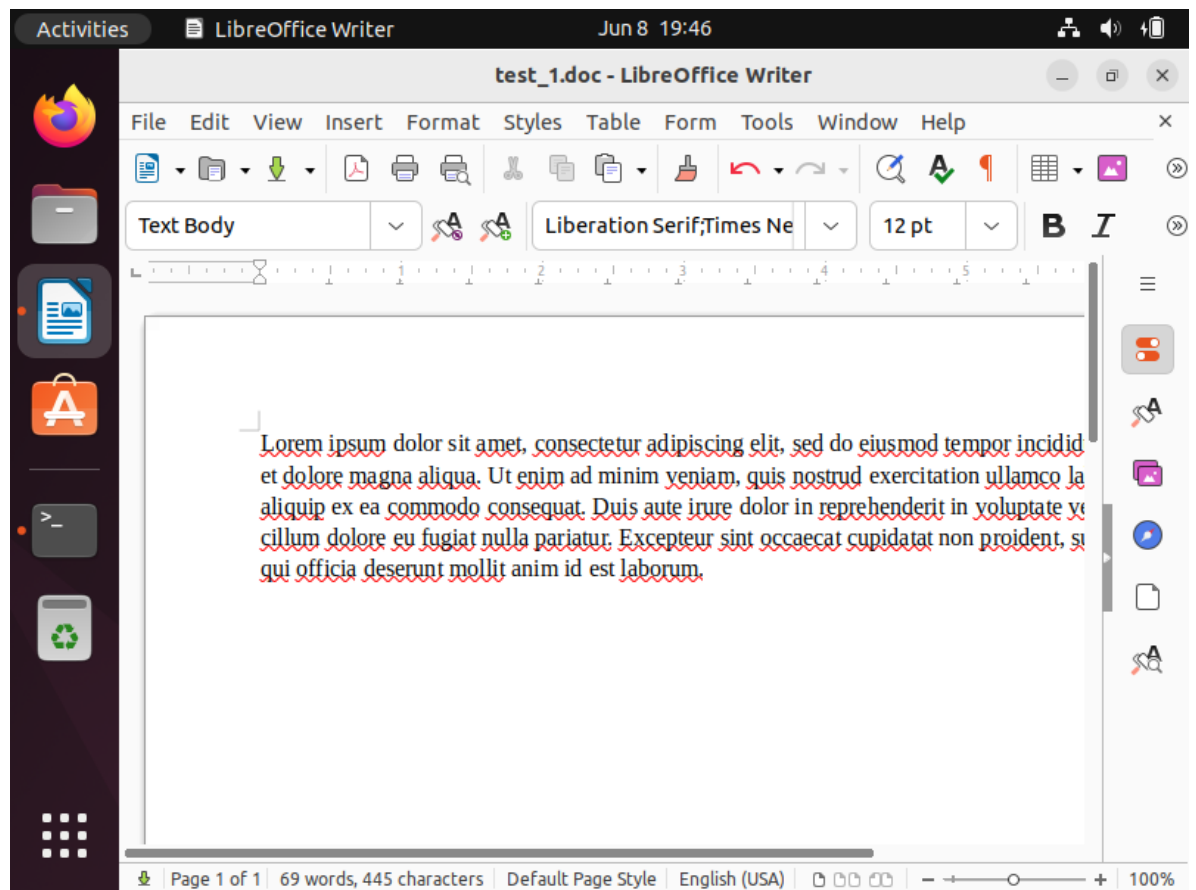
e. Analisis

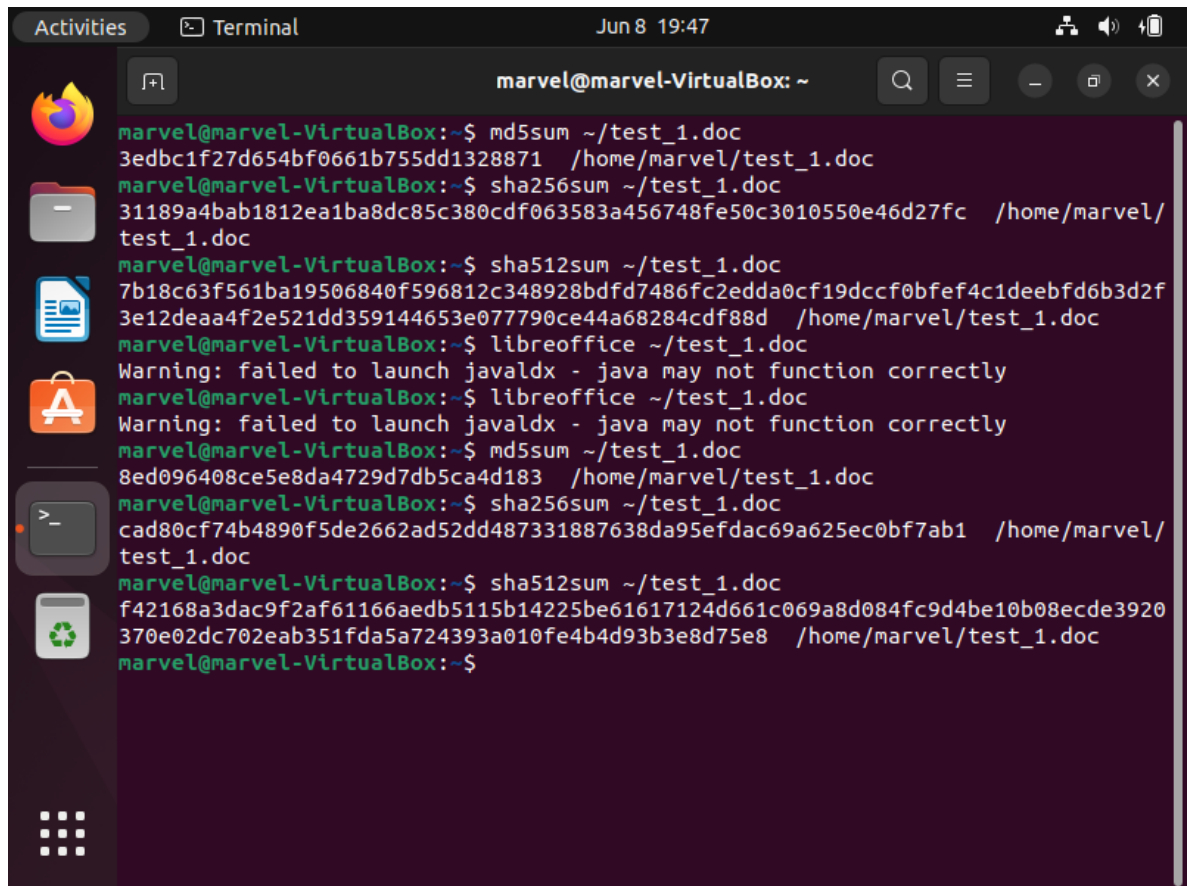
Jawaban:



```
Activities Terminal Jun 8 19:45
marvel@marvel-VirtualBox: ~
marvel@marvel-VirtualBox:~$ md5sum ~/test_1.doc
3edbc1f27d654bf0661b755dd1328871 /home/marvel/test_1.doc
marvel@marvel-VirtualBox:~$ sha256sum ~/test_1.doc
31189a4bab1812ea1ba8dc85c380cdf063583a456748fe50c3010550e46d27fc /home/marvel/
test_1.doc
marvel@marvel-VirtualBox:~$ sha512sum ~/test_1.doc
7b18c63f561ba19506840f596812c348928bdfd7486fc2edda0cf19dccf0bfef4c1deebfd6b3d2f
3e12deaa4f2e521dd359144653e077790ce44a68284cdf88d /home/marvel/test_1.doc
marvel@marvel-VirtualBox:~$
```







```
marvel@marvel-VirtualBox: ~$ md5sum ~/test_1.doc
3edbc1f27d654bf0661b755dd1328871  /home/marvel/test_1.doc
marvel@marvel-VirtualBox: ~$ sha256sum ~/test_1.doc
31189a4bab1812ea1ba8dc85c380cdf063583a456748fe50c3010550e46d27fc  /home/marvel/
test_1.doc
marvel@marvel-VirtualBox: ~$ sha512sum ~/test_1.doc
7b18c63f561ba19506840f596812c348928bdfd7486fc2edda0cf19dccf0bfef4c1deebfd6b3d2f
3e12deaa4f2e521dd359144653e077790ce44a68284cdf88d  /home/marvel/test_1.doc
marvel@marvel-VirtualBox: ~$ libreoffice ~/test_1.doc
Warning: failed to launch javaldx - java may not function correctly
marvel@marvel-VirtualBox: ~$ libreoffice ~/test_1.doc
Warning: failed to launch javaldx - java may not function correctly
marvel@marvel-VirtualBox: ~$ md5sum ~/test_1.doc
8ed096408ce5e8da4729d7db5ca4d183  /home/marvel/test_1.doc
marvel@marvel-VirtualBox: ~$ sha256sum ~/test_1.doc
cad80cf74b4890f5de2662ad52dd487331887638da95efdac69a625ec0bf7ab1  /home/marvel/
test_1.doc
marvel@marvel-VirtualBox: ~$ sha512sum ~/test_1.doc
f42168a3dac9f2af61166aedb5115b14225be61617124d661c069a8d084fc9d4be10b08ecde3920
370e02dc702eab351fda5a724393a010fe4b4d93b3e8d75e8  /home/marvel/test_1.doc
marvel@marvel-VirtualBox: ~$
```

Langkah-langkah:

- a. Download file test_1.doc
- b. Hash awal

```
md5sum ~/test_1.doc
sha256sum ~/test_1.doc
sha512sum ~/test_1.doc
```
- c.
 - i. Buka lagi, ketik abcdef, save
libreoffice ~/test_1.doc
Ketik abcdef → Ctrl+S → Keep Current Format
 - ii. Hapus abcdef, save lagi
Hapus tulisan abcdef → Ctrl+S
- d. Hash lagi setelah semua perubahan

```
md5sum ~/test_1.doc
sha256sum ~/test_1.doc
sha512sum ~/test_1.doc
```
- e. Analisis

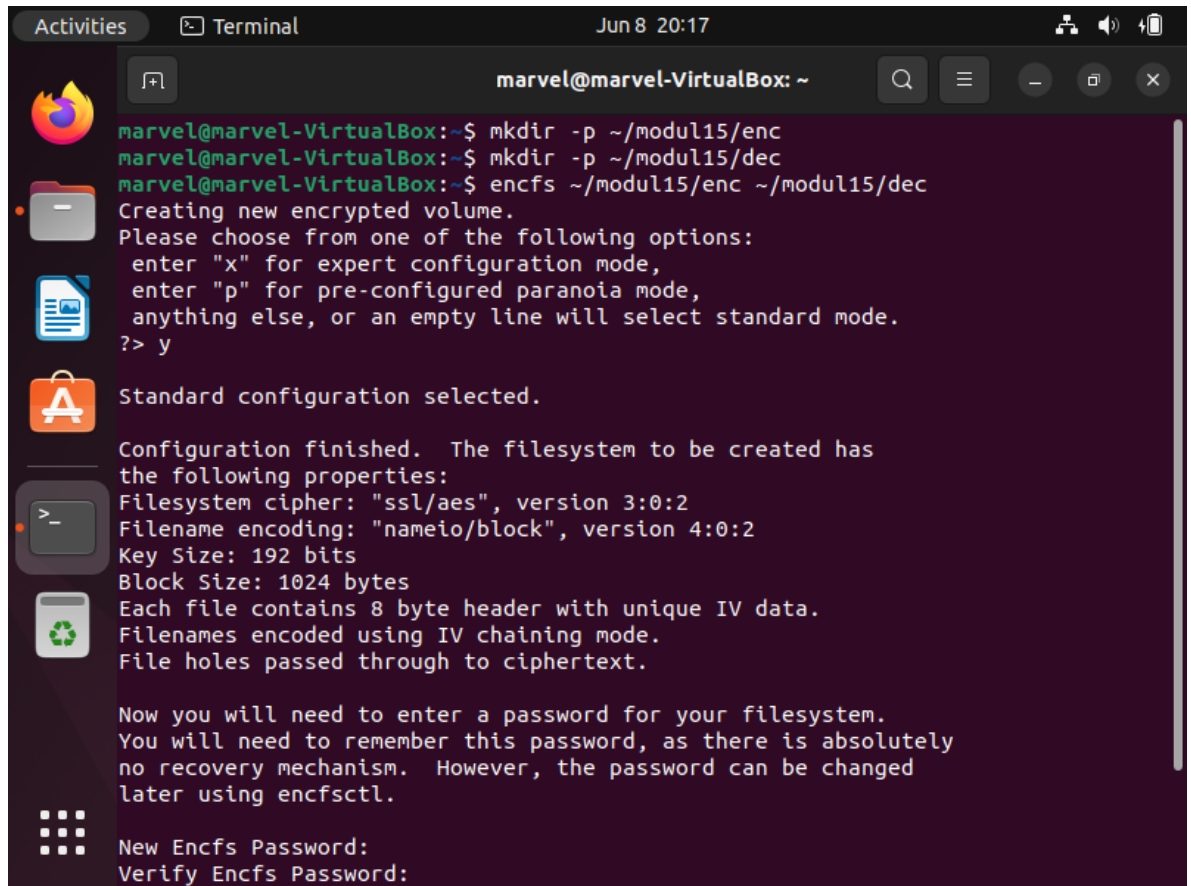
Hash test_1.doc berubah meskipun isi teks dikembalikan seperti semula. Hal ini terjadi karena file .doc menyimpan metadata (waktu edit, jumlah simpan,

revisi, dll) yang ikut berubah setiap kali file dibuka dan disimpan, sehingga hash-nya pun berbeda meskipun konten teksnya sama.

4. Konfidensialitas: encfs

- Buat folder enc (terenkripsi) dan dec (dekripsi), lalu jalankan encfs
- Copy 3 file ke folder_normal (dec), lalu amati folder_terenkripsi (enc)
- Hapus salah satu file di enc, amati dec
- Umount dec, amati kedua folder
- Buat folder_sembarang, mount enc ke sana

Jawaban:



```
marvel@marvel-VirtualBox: ~$ mkdir -p ~/modul15/enc
marvel@marvel-VirtualBox: ~$ mkdir -p ~/modul15/dec
marvel@marvel-VirtualBox: ~$ encfs ~/modul15/enc ~/modul15/dec
Creating new encrypted volume.
Please choose from one of the following options:
  enter "x" for expert configuration mode,
  enter "p" for pre-configured paranoia mode,
  anything else, or an empty line will select standard mode.
?> y

Standard configuration selected.

Configuration finished. The filesystem to be created has
the following properties:
Filesystem cipher: "ssl/aes", version 3:0:2
Filename encoding: "nameio/block", version 4:0:2
Key Size: 192 bits
Block Size: 1024 bytes
Each file contains 8 byte header with unique IV data.
Filenames encoded using IV chaining mode.
File holes passed through to ciphertext.

Now you will need to enter a password for your filesystem.
You will need to remember this password, as there is absolutely
no recovery mechanism. However, the password can be changed
later using encfsctl.

New Encfs Password:
Verify Encfs Password:
```


Activities Terminal Jun 8 20:19

marvel@marvel-VirtualBox: ~

anything else, or an empty line will select standard mode.
?> y

Standard configuration selected.

Configuration finished. The filesystem to be created has the following properties:
Filesystem cipher: "ssl/aes", version 3:0:2
Filename encoding: "nameio/block", version 4:0:2
Key Size: 192 bits
Block Size: 1024 bytes
Each file contains 8 byte header with unique IV data.
Filenames encoded using IV chaining mode.
File holes passed through to ciphertext.

Now you will need to enter a password for your filesystem.
You will need to remember this password, as there is absolutely no recovery mechanism. However, the password can be changed later using encfstl.

New Encfs Password:
Verify Encfs Password:

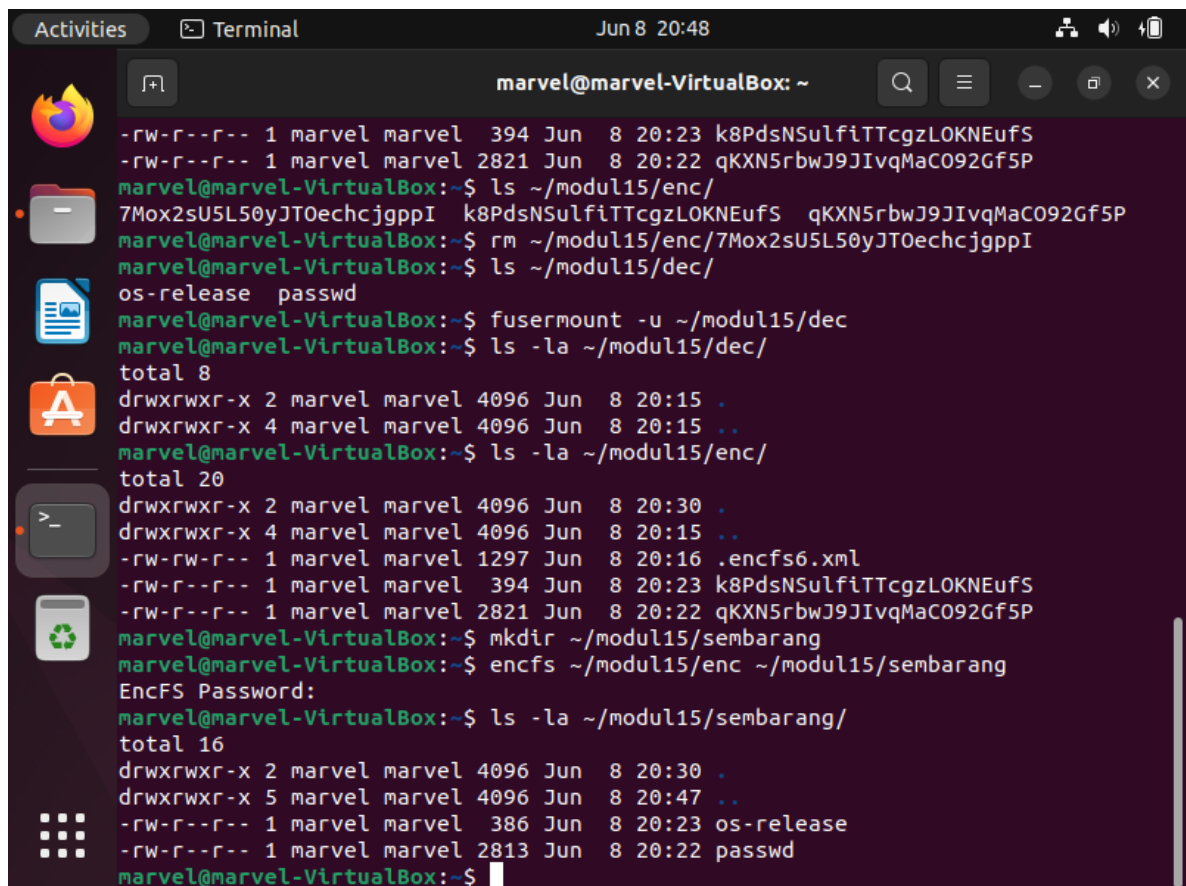
marvel@marvel-VirtualBox:~\$
marvel@marvel-VirtualBox:~\$
marvel@marvel-VirtualBox:~\$ ls ~/modul15/
dec enc
marvel@marvel-VirtualBox:~\$

```
Activities Terminal Jun 8 20:24
marvel@marvel-VirtualBox: ~
later using encfsctl.
New Encfs Password:
Verify Encfs Password:

marvel@marvel-VirtualBox:~$
marvel@marvel-VirtualBox:~$
marvel@marvel-VirtualBox:~$ ls ~/modul15/
dec enc
marvel@marvel-VirtualBox:~$ cp /etc/passwd ~/modul15/dec/
marvel@marvel-VirtualBox:~$ cp /etc/hostname ~/modul15/dec/
marvel@marvel-VirtualBox:~$ cp /etc/os-release ~/modul15/dec/
marvel@marvel-VirtualBox:~$ ls -la ~/modul15/dec/
total 20
drwxrwxr-x 2 marvel marvel 4096 Jun  8 20:23 .
drwxrwxr-x 4 marvel marvel 4096 Jun  8 20:15 ..
-rw-r--r-- 1 marvel marvel  18 Jun  8 20:23 hostname
-rw-r--r-- 1 marvel marvel  386 Jun  8 20:23 os-release
-rw-r--r-- 1 marvel marvel 2813 Jun  8 20:22 passwd
marvel@marvel-VirtualBox:~$ ls -la ~/modul15/enc/
total 24
drwxrwxr-x 2 marvel marvel 4096 Jun  8 20:23 .
drwxrwxr-x 4 marvel marvel 4096 Jun  8 20:15 ..
-rw-r--r-- 1 marvel marvel  26 Jun  8 20:23 7Mox2sU5L50yJT0echcjgppI
-rw-rw-r-- 1 marvel marvel 1297 Jun  8 20:16 .encfs6.xml
-rw-r--r-- 1 marvel marvel  394 Jun  8 20:23 k8PdsNSulfiTTcgzLOKNEufS
-rw-r--r-- 1 marvel marvel 2821 Jun  8 20:22 qKXN5rbwJ9JIvqMaC092Gf5P
marvel@marvel-VirtualBox:~$
```

```
Activities Terminal Jun 8 20:30
marvel@marvel-VirtualBox: ~
marvel@marvel-VirtualBox:~$
marvel@marvel-VirtualBox:~$ ls ~/modul15/
dec enc
marvel@marvel-VirtualBox:~$ cp /etc/passwd ~/modul15/dec/
marvel@marvel-VirtualBox:~$ cp /etc/hostname ~/modul15/dec/
marvel@marvel-VirtualBox:~$ cp /etc/os-release ~/modul15/dec/
marvel@marvel-VirtualBox:~$ ls -la ~/modul15/dec/
total 20
drwxrwxr-x 2 marvel marvel 4096 Jun  8 20:23 .
drwxrwxr-x 4 marvel marvel 4096 Jun  8 20:15 ..
-rw-r--r-- 1 marvel marvel  18 Jun  8 20:23 hostname
-rw-r--r-- 1 marvel marvel  386 Jun  8 20:23 os-release
-rw-r--r-- 1 marvel marvel 2813 Jun  8 20:22 passwd
marvel@marvel-VirtualBox:~$ ls -la ~/modul15/enc/
total 24
drwxrwxr-x 2 marvel marvel 4096 Jun  8 20:23 .
drwxrwxr-x 4 marvel marvel 4096 Jun  8 20:15 ..
-rw-r--r-- 1 marvel marvel  26 Jun  8 20:23 7Mox2sU5L50yJT0echcjgppI
-rw-rw-r-- 1 marvel marvel 1297 Jun  8 20:16 .encfs6.xml
-rw-r--r-- 1 marvel marvel  394 Jun  8 20:23 k8PdsNSulfiTTcgzLOKNEufS
-rw-r--r-- 1 marvel marvel 2821 Jun  8 20:22 qKXN5rbwJ9JIVqMaC092Gf5P
marvel@marvel-VirtualBox:~$ ls ~/modul15/enc/
7Mox2sU5L50yJT0echcjgppI k8PdsNSulfiTTcgzLOKNEufS qKXN5rbwJ9JIVqMaC092Gf5P
marvel@marvel-VirtualBox:~$ rm ~/modul15/enc/7Mox2sU5L50yJT0echcjgppI
marvel@marvel-VirtualBox:~$ ls ~/modul15/dec/
os-release passwd
marvel@marvel-VirtualBox:~$
```

```
Activities Terminal Jun 8 20:45
marvel@marvel-VirtualBox: ~
-rw-r--r-- 1 marvel marvel 18 Jun 8 20:23 hostname
-rw-r--r-- 1 marvel marvel 386 Jun 8 20:23 os-release
-rw-r--r-- 1 marvel marvel 2813 Jun 8 20:22 passwd
marvel@marvel-VirtualBox:~$ ls -la ~/modul15/enc/
total 24
drwxrwxr-x 2 marvel marvel 4096 Jun 8 20:23 .
drwxrwxr-x 4 marvel marvel 4096 Jun 8 20:15 ..
-rw-r--r-- 1 marvel marvel 26 Jun 8 20:23 7Mox2sU5L50yJT0echcjgppI
-rw-rw-r-- 1 marvel marvel 1297 Jun 8 20:16 .encfs6.xml
-rw-r--r-- 1 marvel marvel 394 Jun 8 20:23 k8PdsNSulfiTTcgzLOKNEufS
-rw-r--r-- 1 marvel marvel 2821 Jun 8 20:22 qKXN5rbwJ9JivqMaC092Gf5P
marvel@marvel-VirtualBox:~$ ls ~/modul15/enc/
7Mox2sU5L50yJT0echcjgppI k8PdsNSulfiTTcgzLOKNEufS qKXN5rbwJ9JivqMaC092Gf5P
marvel@marvel-VirtualBox:~$ rm ~/modul15/enc/7Mox2sU5L50yJT0echcjgppI
marvel@marvel-VirtualBox:~$ ls ~/modul15/dec/
os-release passwd
marvel@marvel-VirtualBox:~$ fusermount -u ~/modul15/dec
marvel@marvel-VirtualBox:~$ ls -la ~/modul15/dec/
total 8
drwxrwxr-x 2 marvel marvel 4096 Jun 8 20:15 .
drwxrwxr-x 4 marvel marvel 4096 Jun 8 20:15 ..
marvel@marvel-VirtualBox:~$ ls -la ~/modul15/enc/
total 20
drwxrwxr-x 2 marvel marvel 4096 Jun 8 20:30 .
drwxrwxr-x 4 marvel marvel 4096 Jun 8 20:15 ..
-rw-rw-r-- 1 marvel marvel 1297 Jun 8 20:16 .encfs6.xml
-rw-r--r-- 1 marvel marvel 394 Jun 8 20:23 k8PdsNSulfiTTcgzLOKNEufS
-rw-r--r-- 1 marvel marvel 2821 Jun 8 20:22 qKXN5rbwJ9JivqMaC092Gf5P
marvel@marvel-VirtualBox:~$
```



```
marvel@marvel-VirtualBox: ~  
-rw-r--r-- 1 marvel marvel 394 Jun 8 20:23 k8PdsNSulfiTTcgzLOKNEufS  
-rw-r--r-- 1 marvel marvel 2821 Jun 8 20:22 qKXN5rbwJ9JivqMaC092Gf5P  
marvel@marvel-VirtualBox:~$ ls ~/modul15/enc/  
7Mox2sU5L50yJTOechcjgppI k8PdsNSulfiTTcgzLOKNEufS qKXN5rbwJ9JivqMaC092Gf5P  
marvel@marvel-VirtualBox:~$ rm ~/modul15/enc/7Mox2sU5L50yJTOechcjgppI  
marvel@marvel-VirtualBox:~$ ls ~/modul15/dec/  
os-release passwd  
marvel@marvel-VirtualBox:~$ fusermount -u ~/modul15/dec  
marvel@marvel-VirtualBox:~$ ls -la ~/modul15/dec/  
total 8  
drwxrwxr-x 2 marvel marvel 4096 Jun 8 20:15 .  
drwxrwxr-x 4 marvel marvel 4096 Jun 8 20:15 ..  
marvel@marvel-VirtualBox:~$ ls -la ~/modul15/enc/  
total 20  
drwxrwxr-x 2 marvel marvel 4096 Jun 8 20:30 .  
drwxrwxr-x 4 marvel marvel 4096 Jun 8 20:15 ..  
-rw-rw-r-- 1 marvel marvel 1297 Jun 8 20:16 .encfs6.xml  
-rw-r--r-- 1 marvel marvel 394 Jun 8 20:23 k8PdsNSulfiTTcgzLOKNEufS  
-rw-r--r-- 1 marvel marvel 2821 Jun 8 20:22 qKXN5rbwJ9JivqMaC092Gf5P  
marvel@marvel-VirtualBox:~$ mkdir ~/modul15/sembarang  
marvel@marvel-VirtualBox:~$ encfs ~/modul15/enc ~/modul15/sembarang  
EncFS Password:  
marvel@marvel-VirtualBox:~$ ls -la ~/modul15/sembarang/  
total 16  
drwxrwxr-x 2 marvel marvel 4096 Jun 8 20:30 .  
drwxrwxr-x 5 marvel marvel 4096 Jun 8 20:47 ..  
-rw-r--r-- 1 marvel marvel 386 Jun 8 20:23 os-release  
-rw-r--r-- 1 marvel marvel 2813 Jun 8 20:22 passwd  
marvel@marvel-VirtualBox:~$
```

- Buat folder enc (terenkripsi) dan dec (dekripsi), lalu jalankan encfs:
`mkdir -p ~/modul15/enc ~/modul15/dec`
`encfs ~/modul15/enc ~/modul15/dec`
Pilih y → enter → buat password. Muncul 2 folder enc dan dec. Buat file dengan perintah `nano countdown_1.sh` lalu isi script dengan input user.
- Copy 3 file ke folder_normal (dec), lalu amati folder_terenkripsi (enc):
`cp /etc/passwd /etc/hostname /etc/os-release ~/modul15/dec/`
`ls -la ~/modul15/enc/`
Observasi: File di dec terbaca normal, sedangkan di enc nama filenya terenkripsi (acak).
- Hapus salah satu file di enc, amati dec:
`rm ~/modul15/enc/7Mox2sU5L50yJTOechcjgppI`
`ls ~/modul15/dec/`
Observasi: File yang dihapus dari enc otomatis hilang dari dec juga.
- Umount dec, amati kedua folder:
`fusermount -u ~/modul15/dec`
`ls -la ~/modul15/dec/`
`ls -la ~/modul15/enc/`

Observasi: Folder dec kosong, folder enc masih ada file terenkripsi.

- e. Buat folder `_sembarang`, mount enc ke sana:

`mkdir ~/modul15/sembarang`

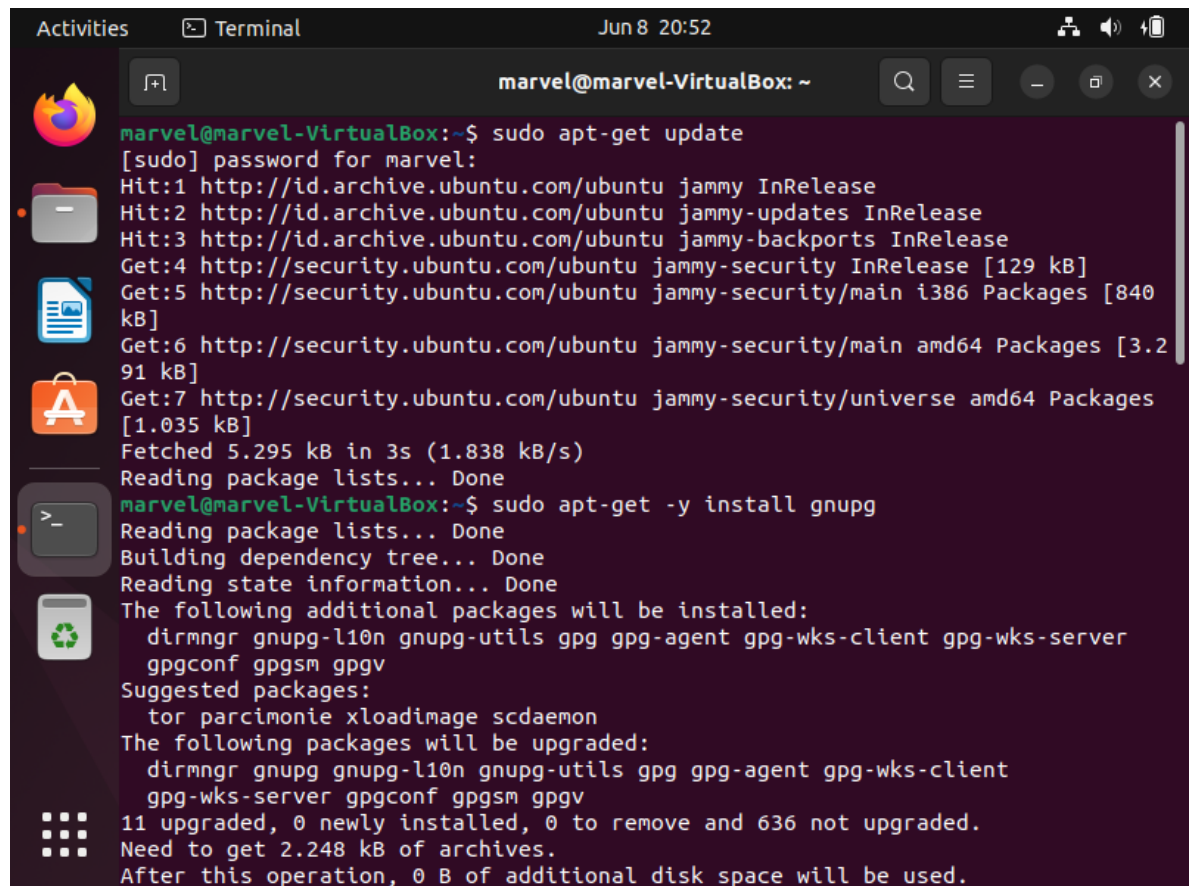
`encfs ~/modul15/enc ~/modul15/sembarang`

Observasi: Setelah memasukkan password yang sama, file di `sembarang` terdekripsi dan terbaca normal sama seperti `dec` sebelumnya.

5. Konfidensialitas: gpg

- Install GPG dan buat keypair
- Cek daftar key dan fingerprint
- Export public key dan rename sesuai NIM
- Mengimport kunci publik orang lain
- Enkripsi pesan
- Enkripsi file untuk teman Anda, unggah hasilnya (.asc), lalu unduh dan dekripsi file yang ditujukan untuk Anda menggunakan GPG.

Jawaban:



```
Activities Terminal Jun 8 20:52
marvel@marvel-VirtualBox: ~
marvel@marvel-VirtualBox:~$ sudo apt-get update
[sudo] password for marvel:
Hit:1 http://id.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://id.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://id.archive.ubuntu.com/ubuntu jammy-backports InRelease
Get:4 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:5 http://security.ubuntu.com/ubuntu jammy-security/main i386 Packages [840
kB]
Get:6 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3.2
91 kB]
Get:7 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages
[1.035 kB]
Fetched 5.295 kB in 3s (1.838 kB/s)
Reading package lists... Done
marvel@marvel-VirtualBox:~$ sudo apt-get -y install gnupg
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  dirmngr gnupg-l10n gnupg-utils gpg gpg-agent gpg-wks-client gpg-wks-server
  gpgconf gpgsm gpgv
Suggested packages:
  tor parcimonie xloadimage sdcdaemon
The following packages will be upgraded:
  dirmngr gnupg gnupg-l10n gnupg-utils gpg gpg-agent gpg-wks-client
  gpg-wks-server gpgconf gpgsm gpgv
11 upgraded, 0 newly installed, 0 to remove and 636 not upgraded.
Need to get 2.248 kB of archives.
After this operation, 0 B of additional disk space will be used.
```

Activities Terminal Jun 8 20:54

marvel@marvel-VirtualBox: ~

```
marvel@marvel-VirtualBox:~$ gpg --gen-key
gpg (GnuPG) 2.2.27; Copyright (C) 2021 Free Software Foundation, Inc.
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Note: Use "gpg --full-generate-key" for a full featured key generation dialog.

GnuPG needs to construct a user ID to identify your key.

Real name: Marvel
Email address: marvelsanjaya7@gmail.com
You selected this USER-ID:
    "Marvel <marvelsanjaya7@gmail.com>"

Change (N)ame, (E)mail, or (O)kay/(Q)uit? O
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: key 24F56B1FC36542CC marked as ultimately trusted
gpg: directory '/home/marvel/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/marvel/.gnupg/openpgp-revocs.d/64A
820D4128E2304925F6A5124F56B1FC36542CC.rev'
public and secret key created and signed.
```



```
Activities  Terminal  Jun 8 20:54  marvel@marvel-VirtualBox: ~

GnuPG needs to construct a user ID to identify your key.

Real name: Marvel
Email address: marvelsanjaya7@gmail.com
You selected this USER-ID:
    "Marvel <marvelsanjaya7@gmail.com>"

Change (N)ame, (E)mail, or (O)kay/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: key 24F56B1FC36542CC marked as ultimately trusted
gpg: directory '/home/marvel/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/marvel/.gnupg/openpgp-revocs.d/64A
820D4128E2304925F6A5124F56B1FC36542CC.rev'
public and secret key created and signed.

pub  rsa3072 2026-06-08 [SC] [expires: 2028-06-07]
     64A820D4128E2304925F6A5124F56B1FC36542CC
uid                               Marvel <marvelsanjaya7@gmail.com>
sub  rsa3072 2026-06-08 [E] [expires: 2028-06-07]

marvel@marvel-VirtualBox:~$
```

Activities Terminal Jun 8 20:55

marvel@marvel-VirtualBox: ~

Change (N)ame, (E)mail, or (O)kay/(Q)uit? O

We need to generate a lot of random bytes. It is a good idea to perform some other action (type on the keyboard, move the mouse, utilize the disks) during the prime generation; this gives the random number generator a better chance to gain enough entropy.

We need to generate a lot of random bytes. It is a good idea to perform some other action (type on the keyboard, move the mouse, utilize the disks) during the prime generation; this gives the random number generator a better chance to gain enough entropy.

gpg: key 24F56B1FC36542CC marked as ultimately trusted

gpg: directory '/home/marvel/.gnupg/openpgp-revocs.d' created

gpg: revocation certificate stored as '/home/marvel/.gnupg/openpgp-revocs.d/64A820D4128E2304925F6A5124F56B1FC36542CC.rev'

public and secret key created and signed.

pub rsa3072 2026-06-08 [SC] [expires: 2028-06-07]
64A820D4128E2304925F6A5124F56B1FC36542CC

uid Marvel <marvelsanjaya7@gmail.com>

sub rsa3072 2026-06-08 [E] [expires: 2028-06-07]

marvel@marvel-VirtualBox:~\$ gpg --list-keys
/home/marvel/.gnupg/pubring.kbx

pub rsa3072 2026-06-08 [SC] [expires: 2028-06-07]
64A820D4128E2304925F6A5124F56B1FC36542CC

uid [ultimate] Marvel <marvelsanjaya7@gmail.com>

sub rsa3072 2026-06-08 [E] [expires: 2028-06-07]

marvel@marvel-VirtualBox:~\$

Activities Terminal Jun 8 20:56

marvel@marvel-VirtualBox: ~

some other action (type on the keyboard, move the mouse, utilize the disks) during the prime generation; this gives the random number generator a better chance to gain enough entropy.
gpg: key 24F56B1FC36542CC marked as ultimately trusted
gpg: directory '/home/marvel/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/marvel/.gnupg/openpgp-revocs.d/64A820D4128E2304925F6A5124F56B1FC36542CC.rev'
public and secret key created and signed.

pub rsa3072 2026-06-08 [SC] [expires: 2028-06-07]
64A820D4128E2304925F6A5124F56B1FC36542CC
uid Marvel <marvelsanjaya7@gmail.com>
sub rsa3072 2026-06-08 [E] [expires: 2028-06-07]

marvel@marvel-VirtualBox:~\$ gpg --list-keys /home/marvel/.gnupg/pubring.kbx

pub rsa3072 2026-06-08 [SC] [expires: 2028-06-07]
64A820D4128E2304925F6A5124F56B1FC36542CC
uid [ultimate] Marvel <marvelsanjaya7@gmail.com>
sub rsa3072 2026-06-08 [E] [expires: 2028-06-07]

marvel@marvel-VirtualBox:~\$ gpg --fingerprint marvelsanjaya7@gmail.com
pub rsa3072 2026-06-08 [SC] [expires: 2028-06-07]
64A8 20D4 128E 2304 925F 6A51 24F5 6B1F C365 42CC
uid [ultimate] Marvel <marvelsanjaya7@gmail.com>
sub rsa3072 2026-06-08 [E] [expires: 2028-06-07]

marvel@marvel-VirtualBox:~\$

```
Activities Terminal Jun 8 20:59
marvel@marvel-VirtualBox: ~

pub rsa3072 2026-06-08 [SC] [expires: 2028-06-07]
64A820D4128E2304925F6A5124F56B1FC36542CC
uid Marvel <marvelsanjaya7@gmail.com>
sub rsa3072 2026-06-08 [E] [expires: 2028-06-07]

marvel@marvel-VirtualBox:~$ gpg --list-keys
/home/marvel/.gnupg/pubring.kbx
-----
pub rsa3072 2026-06-08 [SC] [expires: 2028-06-07]
64A820D4128E2304925F6A5124F56B1FC36542CC
uid [ultimate] Marvel <marvelsanjaya7@gmail.com>
sub rsa3072 2026-06-08 [E] [expires: 2028-06-07]

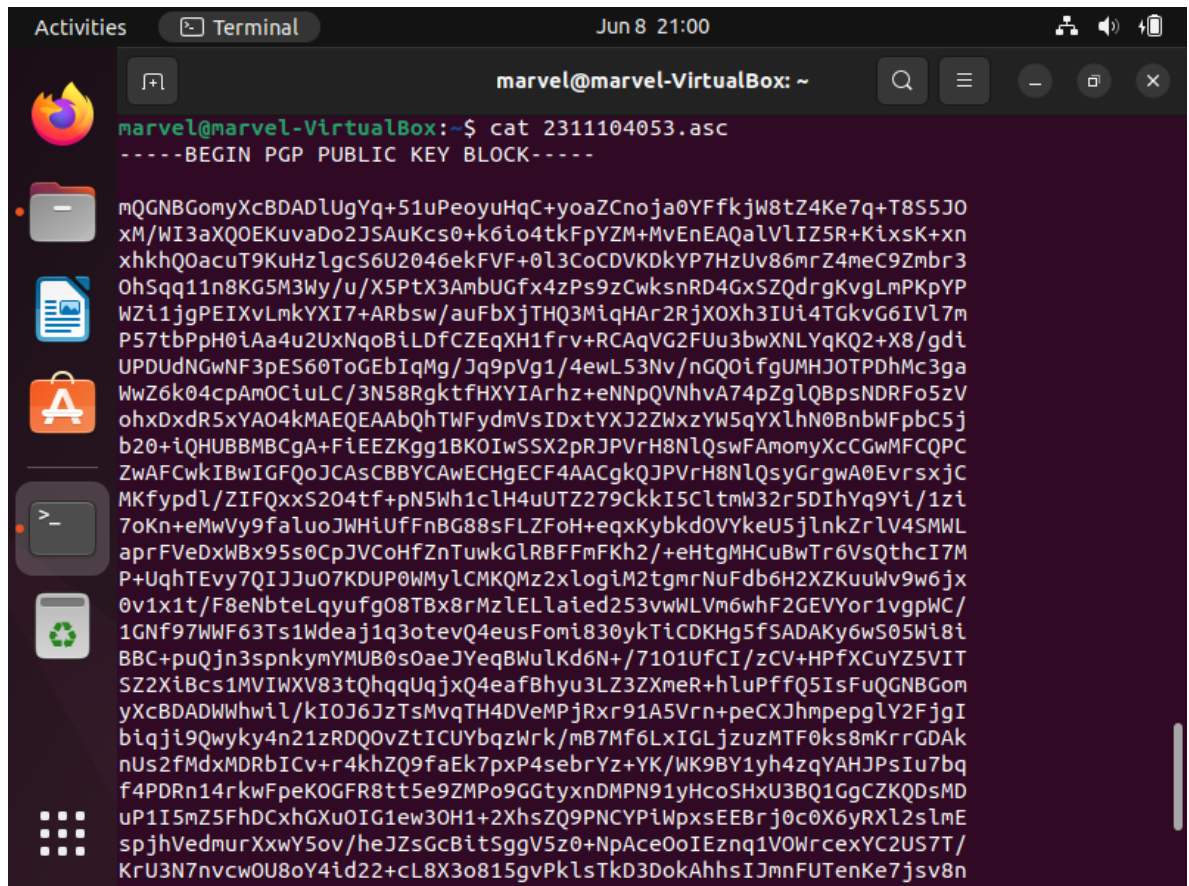
marvel@marvel-VirtualBox:~$ gpg --fingerprint marvelsanjaya7@gmail.com
pub rsa3072 2026-06-08 [SC] [expires: 2028-06-07]
64A8 20D4 128E 2304 925F 6A51 24F5 6B1F C365 42CC
uid [ultimate] Marvel <marvelsanjaya7@gmail.com>
sub rsa3072 2026-06-08 [E] [expires: 2028-06-07]

marvel@marvel-VirtualBox:~$ gpg --armor --export marvelsanjaya7@gmail.com > mypublic_key.asc
marvel@marvel-VirtualBox:~$ ls
Desktop Encrypt modul15 Pictures Templates
Documents file_0.txt Music Public test_1.doc
Downloads modul14 mypublic_key.asc snap Videos
marvel@marvel-VirtualBox:~$ mv mypublic-key.asc 2311104053.asc
mv: cannot stat 'mypublic-key.asc': No such file or directory
marvel@marvel-VirtualBox:~$ mv mypublic_key.asc 2311104053.asc
marvel@marvel-VirtualBox:~$
```

```
Activities Terminal Jun 8 21:00
marvel@marvel-VirtualBox: ~
marvel@marvel-VirtualBox:~$ gpg --list-keys
/home/marvel/.gnupg/pubring.kbx
-----
pub  rsa3072 2026-06-08 [SC] [expires: 2028-06-07]
    64A820D4128E2304925F6A5124F56B1FC36542CC
uid  [ultimate] Marvel <marvelsanjaya7@gmail.com>
sub  rsa3072 2026-06-08 [E] [expires: 2028-06-07]

marvel@marvel-VirtualBox:~$ gpg --fingerprint marvelsanjaya7@gmail.com
pub  rsa3072 2026-06-08 [SC] [expires: 2028-06-07]
    64A8 20D4 128E 2304 925F 6A51 24F5 6B1F C365 42CC
uid  [ultimate] Marvel <marvelsanjaya7@gmail.com>
sub  rsa3072 2026-06-08 [E] [expires: 2028-06-07]

marvel@marvel-VirtualBox:~$ gpg --armor --export marvelsanjaya7@gmail.com > mypublic_key.asc
marvel@marvel-VirtualBox:~$ ls
Desktop  Encrypt  modul15  Pictures  Templates
Documents file_0.txt Music    Public    test_1.doc
Downloads modul14  mypublic_key.asc snap      Videos
marvel@marvel-VirtualBox:~$ mv mypublic-key.asc 2311104053.asc
mv: cannot stat 'mypublic-key.asc': No such file or directory
marvel@marvel-VirtualBox:~$ mv mypublic_key.asc 2311104053.asc
marvel@marvel-VirtualBox:~$ ls
2311104053.asc Downloads modul14 Pictures Templates
Desktop      Encrypt    modul15 Public    test_1.doc
Documents    file_0.txt Music     snap      Videos
marvel@marvel-VirtualBox:~$
```



The screenshot shows a terminal window titled 'Terminal' with the date 'Jun 8 21:00'. The prompt is 'marvel@marvel-VirtualBox: ~'. The user has executed the command 'cat 2311104053.asc', which displays a GPG public key block. The key block starts with '-----BEGIN PGP PUBLIC KEY BLOCK-----' and contains a long string of base64-encoded data. The terminal window has a sidebar on the left with icons for Activities, Terminal, Files, Documents, Applications, and Trash.

```
marvel@marvel-VirtualBox:~$ cat 2311104053.asc
-----BEGIN PGP PUBLIC KEY BLOCK-----

mQGNBGomyXcBDADlUgYq+51uPeoyuHqC+yoaZCnoja0YFfkjW8tZ4Ke7q+T8S5JO
xM/WI3aXQ0EKuvaDo2JSAuKcs0+k6io4tkFpYZM+MvEnEAQalVLIZ5R+KixsK+xn
xhkhQ0acuT9KuHzlgcS6U2046ekFVF+0l3CoCDVKdKYP7HzUv86mrZ4meC9Zmbr3
OhSqq11n8KG5M3Wy/u/X5PtX3AmbUGfx4zPs9zCwksnRD4GxSZQdrgKvgLmPKpYP
WZi1jgPEIXvLmkYXI7+ARbsw/auFbXjTHQ3MiQhAR2RjXOXh3IU14TGkvG6IVl7m
P57tbPpH0iAa4u2UxNqoBiLDfCZEqXH1frv+RCAqVG2FUu3bwXNLYqKQ2+X8/gdi
UPDUdNGwNF3pES60ToGEbIqMg/Jq9pVg1/4ewL53Nv/nGQ0ifgUMHJOTPDhMc3ga
WwZ6k04cpAmOCiulC/3N58RgktfHXyIArhz+eNNpQVNHvA74pZglQBpsNDRFo5zV
ohxDxdR5xYA04kMAEQEAAbQhTWfYdmVsIDxtYXJ2ZWxzYW5qYXlnb0BnbWpbc5j
b20+iQHUBBMBcGA+FiEEZKgg1BK0IwSSX2pRJPVrH8NLQswFamomyXcCGwMFCQPC
ZwAFcwkIBwIGFQoJCAACBBYCAWECHgECF4AACgkQJPVrH8NLQsyGrgWA0EvrsxjC
MKfypdl/ZIFQxsS204tf+pN5Wh1clH4uUTZ279CkKi5ClmW32r5DIhYq9Yi/1zi
7oKn+eMwVy9faluoJWHiUfFNBG88sFLZFoH+eqxKybkD0VYkeU5jlnkZrLV4SMWL
aprFVeDxWBx95s0CpJVCohfZnTuwkGLRBFFmFkh2/+eHtgMHCuBwTr6VsQthcI7M
P+UqhTEvy7QIJJu07KDUP0WMyLCMKQMz2xlogiM2tgmrNuFdb6H2XKuuWv9w6jx
0v1x1t/F8eNbteLqyufg08TBx8rMzLELlaied253vwWLVm6whF2GEVYor1vgpWC/
1GNf97WWF63Ts1Wdeaj1q3otevQ4eusFomi830ykTiCDKHg5fSADAKy6wS05Wi8i
BBC+puQjn3spnkymYMUB0s0aeJYeqBWuLkd6N+/7101UfCI/zCV+HPfXCuYZ5VIT
SZ2XiBcs1MVIWV83tQhqqUqjxQ4eafBhyu3LZ3XmeR+hluPffQ5IsFuQGNBGom
yXcBDADWWwhil/kIOJ6JzTsmvqTH4DVeMPjRxr91A5Vrn+peCXJhmpgLY2FjgI
biqji9Qwyky4n21zRDQ0vZtICUybqzWrk/mB7Mf6LxIGLjzuzMTF0ks8mKrrGDAk
nUs2fMdxMDRbICv+r4khZQ9faEk7pxP4sebrYz+YK/WK9BY1yh4zqYAHJPsiU7bq
f4PDRn14rkWfpeKOGFR8tt5e9ZMPo9GGtyxndMPN91yHcoSHxU3BQ1GgCZKQDSMD
uP1I5mZ5FhDCxhGXu0IG1ew30H1+2XhsZQ9PNCYPiWpxsEEBrj0c0X6yRXl2sLmE
spjhVedmurXxwY5ov/heJZsGcBltSggV5z0+NpAce0oIEznq1V0WrcexYC2US7T/
KrU3N7nvcwOU8oY4id22+cL8X3o815gvPklsTkD3DokAhhsIJmnFUTenKe7jsv8n
```

Langkah-langkah:

a. Install GPG dan buat keypair:

```
sudo apt-get install gnupg
```

```
gpg --gen-key
```

Nama: Marvel, Email: marvelsanjaya7@gmail.com. Key ID:

64A820D4128E2304925F6A5124F56B1FC36542CC

b. Cek daftar key dan fingerprint:

```
gpg --list-keys
```

```
gpg --fingerprint marvelsanjaya7@gmail.com
```

Fingerprint: 64A8 20D4 128E 2304 925F 6A51 24F5 6B1F C365 42CC

c. Export public key dan rename sesuai NIM:

```
gpg --armor --export marvelsanjaya7@gmail.com > mypublic_key.asc
```

```
mv mypublic_key.asc 2311104053.asc
```

File 2311104053.asc berhasil dibuat

5d, 5e, 5f — Belum bisa dikerjakan karena menunggu teman untuk tukar public key dan enkripsi/dekripsi pesan.